Fingerprints

01 What made the curved lines on the glass?



A fingertip can leave a map of its raised ridges.

Look closer. Look for the pale ridges and the darker spaces between them.

A little explanation. A fingertip has raised lines called friction ridges, with lower grooves between them. When ridge tops touch a surface, sweat and natural skin oils can leave a faint impression of their paths.

Try this. Tilt a clean drinking glass toward a lamp and look for a faint print. Keep the glass on the table while you look.

02 Which way do the ridges flow?



Loops, whorls, and arches describe the broad ridge flow.

Look closer. Trace one curve in each large pattern card with your eyes.

A little explanation. Many fingerprints can be grouped by their broad ridge flow. A loop curves back toward one side, a whorl circles around a center, and an arch rises and crosses from one side to the other.

Try this. Draw one loop, one circle-like whorl, and one hill-shaped arch on paper.

03 Can two loops still be different?



The same broad pattern can hold different tiny details.

Look closer. Find a ridge that ends and another that splits into two paths.

A little explanation. The big pattern family is only a first clue. Along the ridges are smaller details: a line may end, split, shorten, or bend. Two prints can both be loops while their tiny details appear in different places.

Try this. On two drawn loops, add endings and forks in different places. Compare your maps.

04 Why does a print have dark lines and pale gaps?



Ridge tops transfer ink; lower grooves leave pale gaps.

Look closer. Compare the inked fingertip with the print it leaves on paper.

A little explanation. Washable ink touches the raised ridge tops. When the fingertip presses paper, those tops transfer dark lines while the lower grooves leave pale spaces. The print is a flat impression of a textured fingertip.

Try this. Use a non-toxic washable stamp pad on one fingertip, press once on paper, and wash your hands. Keep the paper at home.